

What is JavaScript.?

JavaScript is a **high-level, interpreted (now JIT-compiled), dynamically typed programming language** primarily used to build **interactive, event-driven web applications**.

Q: Is JavaScript compiled or interpreted?

It is **not purely interpreted** anymore.

Modern engines (like V8) follow this process:

JavaScript uses Just-In-Time (JIT) compilation, meaning the code is parsed and compiled into machine code at runtime before execution.

Q: What is V8?

Google's JS Engine written in C++.

Q: Why Doesn't JavaScript Need Manual Memory Allocation?

JavaScript doesn't require manual memory allocation because it uses automatic garbage collection. The engine allocates memory when values are created and frees it when those values become unreachable, typically using a mark-and-sweep algorithm. This allows developers to focus on application logic instead of managing memory manually. When you create values in JavaScript:

```
let user = { name: "Prajwal" };
```

The JavaScript engine automatically:

- 1 Allocates memory** when objects/variables are created.
- 2 Tracks references** to know whether the data is still being used.
- 3 Frees memory automatically** when the data is no longer reachable.

Stack vs Heap (Managed Automatically)

- **Stack** → Stores primitives & function calls (auto cleaned after execution)
- **Heap** → Stores objects, arrays, closures (GC cleans unused data)

Abstract Syntax Tree(AST) is a hierarchical representation of source code generated by the parser. It captures the logical structure of the program, allowing the JavaScript engine and developer tools to analyze, optimize, and execute the code efficiently.

How javascript code runs?

Code → Parsing → AST → Compilation → Execution